Professional Online Academy Vu Topper RM

MTH-601 Operations Research
Update MCQ'S Final Term
By Vu Topper RM

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Question No:1 **(Marks:1)** **Vu-Topper RM**

How is Vogel's Approximation Method (VAM) different from the least cost method (LCM)?

A. VAM does not consider penalties
B. VAM considers penalties for wrong allocations
C. VAM does not consider unit costs
D. VAM starts from the bottom right corner

Question No:2 **(Marks:1)** **Vu-Topper RM**

The Transportation Problem is one of the sub-class of ----------.

A. Inventory Control Problem
B. Stochastic Problem
C. Network Flow Problem
D. Linear Programming Problem

Question No:3 **(Marks:1)** **Vu-Topper RM**

If the cost matrix in an Assignment problem is not square then which the following modification will be made to balance the given problem?

A. Add a dummy row(column) with arbitrary(including negative or positive) cost elements
B. Add a dummy row(column) with positive cost elements
C. Add a dummy row(column) with zero cost elements
D. Add a dummy row(column) with negative cost elements

Question No:4 **(Marks:1)** **Vu-Topper RM**

Suppose that a taxi firm has four taxis (the agents) available, and four customers (the tasks) wishing to be picked up as soon as possible. The firm prides itself on speedy pickups, so for each taxi the cost of picking up a particular customer will depend on the time taken for the taxi to reach the pickup point. This is _____ problem.

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A. transportation
B. unbalanced assignment
C. balanced assignment
D. linear programming

Question No:5 **(Marks:1)** **Vu-Topper RM**

_____ problems are _____ problems

A. Assignment, combinatorial
B. transportation, unbalanced
C. none of these
D. linear programming, simplex

Question No:6 **(Marks:1)** **Vu-Topper RM**

For calculating the penalty cost of a row in Vogel's approximation method, if the smallest element is '3' and other elements are 3, 4, 5 then which of the following will be taken as next highest?

A. 3
B. 5
C. 7
D. 4

Question No:7 **(Marks:1)** **Vu-Topper RM**

To balance Assignment matrix we have to----------.

A. open a Dummy column
B. you cannot balance the assignment matrix
C. open a Dummy row
D. open either a dummy row or column depending on the situation

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Question No:8 **(Marks:1)** **Vu-Topper RM**

Under which of the following condition in the North West Corner Method, after first allocation in cell (1,1), next allocation along the diagonal in cell (2,2) can be made?

A. If in the first row and first column, each resource and sink contain equal units

B. No such condition exist

C. If Total Supply is not equal to Total Demand

D. Total Supply = Total Demand

Question No:9 **(Marks:1)** **Vu-Topper RM**

In North West Corner Rule, we start initially from which of the following cell in the standard transportation table?

A. Upper left

B. Upper right

C. Upper middle

D. From any arbitrary cell in first row

Question No:10 **(Marks:1)** **Vu-Topper RM**

In transportation models, what does the decision variable xij typically represent?

A. The total production capacity at factory i.

B. The total demand at warehouse j.

C. The quantity of goods transported from factory i to warehouse j.

D. The cost of transporting one unit from factory i to warehouse j.

Question No:11 **(Marks:1)** **Vu-Topper RM**

If and are row and column numbers respectively, then the implied cost is given by: Ui Vj Cij

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A. Ui– Vj
B. Ui/Vj
C. Ui + Vj
D. Ui × Vj

Question No:12 **(Marks:1)** **Vu-Topper RM**

For North West Corner method, in the first row and first column, each resource and sink contain '5' units; then after allocating the appropriate amount 'x11' in the cell (1,1), in which of the following cell the next allocation will be zero?

A. Neither in (1,2) nor in (2,1) but in along the diagonal in (2,2)
B. In cell (2,1) only
C. Arbitrarily chosen either in (1,2) or in (2,1)
D. In cell (1,2) only

Question No:13 **(Marks:1)** **Vu-Topper RM**

Which of the following reason to prefer Vogel's approximation method over other two methods?

A. Vogel's method gives the shortest transportation rout
B. Vogel's method allocate minimum amount to the variable with least unit cost
C. Vogel's method gives optimal time plan of transportation
D. Vogel's method gives the feasible solution near to the optimal

Question No:14 **(Marks:1)** **Vu-Topper RM**

The amounts shipped from a dummy source represent shortages at the receiving destinations

True
False

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Question No:15 **(Marks:1)** **Vu-Topper RM**

In the Least Cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?

A. Only column
B. Either of row or column
C. Only row
D. Both row and column

Question No:16 **(Marks:1)** **Vu-Topper RM**

The similarity between Assignment Problem and Transportation problem is----------.

A. both are square matrices
B. both are rectangular matrices
C. both have objective function and non negativity constraints
D. both can be solved by graphical method

Question No:17 **(Marks:1)** **Vu-Topper RM**

While solving an Assignment problem by Hungarian's method, the horizontal and vertical lines drawn to cover all zeros of modified cost matrix must be ---------.

A. must be equal to m.n (where m and n are number of rows and columns)
B. equal to number of rows or columns
C. equal to each other
D. m + n(m and n are number of rows and columns)

Question No:18 **(Marks:1)** **Vu-Topper RM**

If a balanced Transportation problem with 'm' sources and 'n' sinks then degeneracy will arise only if there are less than ---------- independent allocations in the solution.

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A. m–n–4
B. m–n–2
C. m–n–3
D. m–n–1

Question No:19 (Marks:1) Vu-Topper RM

In an Assignment problem, while applying Hungarian's method, if in the final modified matrix any row or column does not have single zero, then which of the following is true?

A. Unique assignment will be made
B. None of the above
C. No assignment can be made
D. Multiple assignments can be made

Question No:20 (Marks:1) Vu-Topper RM

What is done with the row/column after an allocation is made in the LCM?

A. Eliminate the row/column
B. Adjust the demand at the destination
C. Increase the cost of transportation
D. Retain the row/column for future allocations

Question No:21 (Marks:1) Vu-Topper RM

If the total cost of the assignment for all the tasks is equal to the sum of the costs for each agent, then the problem is called _____.

A. optimization
B. simplex
C. linear assignment
D. unbalanced assignment

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Question No:22 **(Marks:1)** **Vu-Topper RM**

In the context of transportation problems, what does it mean for a solution to be non-degenerate?

A. It has no allocations at all.
B. It has the minimum possible number of allocations.
C. It has the maximum possible number of allocations.
D. It has exactly (m + n - 1) allocations in independent positions.

Question No:23 **(Marks:1)** **Vu-Topper RM**

For an unbalanced Transportation problem, if the total demand is MORE than total supply then which of the following is true in order to balance the problem?

A. One constraint will have evacuate
B. One constraint will have to add
C. A dummy source would have to include with supply equal to shortage
D. A dummy sink would have to include with demand equal to the surplus

Question No:24 **(Marks:1)** **Vu-Topper RM**

There always exists a non-degenerate Feasible solution of a Balanced Transportation Problem.

True
False

Question No:25 **(Marks:1)** **Vu-Topper RM**

If a balanced Transportation problem with '7' sources and '6'sinks then its non-degenerate basic feasible solution will have---------- independent allocations.

A. 11

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B. 12
C. 13
D. 10

Question No:26 **(Marks:1)** **Vu-Topper RM**

Despite of minimizing the cost of Transportation, if we want to maximize the profit then which of the following modification would have to do?

A. Subtract '1' from all the cost coefficients in Transportation matrix
B. Divide all the cost coefficients in Transportation matrix by minimum cost coefficient
C. Add '1' from all the cost coefficients in Transportation matrix
D. Multiply '–1' to all the cost coefficients in Transportation matrix

Question No:27 **(Marks:1)** **Vu-Topper RM**

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____ .

A. non of these
B. does not have a solution
C. may have a solution
D. must have a solution

Question No:28 **(Marks:1)** **Vu-Topper RM**

Once an LP problem is converted to a table, then calculation steps would be same for _____ method(s).

A. Big M
B. all of above
C. simplex
D. Two phase

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Question No:29 **(Marks:1)** **Vu-Topper RM**

In two phase method, for the phase-I, if the objective function in terms of artificial variables is not minimized then the given problem has---------.

A. Degeneracy
B. Feasible solution
C. Optimal solution
D. Infeasible solution

Question No:30 **(Marks:1)** **Vu-Topper RM**

The role of artificial variables in the 'Big M' technique is to _____.

A. maximize the objective function
B. minimize the objective function
C. ensure feasibility of the initial basic solution
D. optimize resource allocation

Question No:31 **(Marks:1)** **Vu-Topper RM**

By Simplex method, to maximize 'Z = 2x–9y'of an LP problem, if 'z=0' for the initial iteration then for its next improved solution, which of the following would
be the next entering variable?

A. y<0
B. x>0
C. x<0
D. y>0

Question No:32 **(Marks:1)** **Vu-Topper RM**

If a variable in the Primal is unrestricted in sign, then the corresponding constraint in the dual will be of -------- type and vice versa.

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A. =
B. none of these
C. <=
D. >=

Question No:33 **(Marks:1)** **Vu-Topper RM**
While solving an LP problem by Simplex method, the inclusion of slacks in the constraints' inequalities helps in finding ---------- variables.

A. basic
B. non-basic
C. artificial
D. decision

Question No:34 **(Marks:1)** **Vu-Topper RM**
In two phase method, for the phase-I, the problem has infeasible solution if the minimum value of objective function ----------.

A. is imaginary
B. greater than zero
C. less than zero
D. equal to zero

Question No:35 **(Marks:1)** **Vu-Topper RM**
In the Two Phase Method the objective of Phase 2 is to _____.

A. maximize the artificial variables
B. find the optimum basic solution for the original problem
C. minimize the original objective function
D. maximize the original objective function

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Question No:36 **(Marks:1)** **Vu-Topper RM**

Dual of a Dual of Dual is------.

A. Single dual
B. Dual
C. Double dual
D. Primal

Question No:37 **(Marks:1)** **Vu-Topper RM**

In Big M-method, if the objective function is expressed in terms of non-basic variable then the coefficient of artificial variable in this iteration must be ----
---.

A. one
B. M
C. –M
D. zero

Question No:38 **(Marks:1)** **Vu-Topper RM**

If in a LP problem, the objective is to maximize: z = 2x+3y, with all associated constraints of '=<' type, then in the 1st iteration, z = ---------.

A. 0
B. 5
C. 6
D. 1

Question No:39 **(Marks:1)** **Vu-Topper RM**

The insensitivity of the solution relative to the original decision variables in an LP problem which is solved by M-method is overcome by ----------.

A. Graphical method

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B. Two phase method
C. Duality principle
D. Simplex method

Question No:40 **(Marks:1)** **Vu-Topper RM**
Simplex method can only be applied to problems with two decision variables.
True
False

Question No:41 **(Marks:1)** **Vu-Topper RM**
If the optimal table contains more zero coefficients in the objective row than the number of variables in the basis in simplex procedure then the solution
will _____.
A. not exist
B. indicate an unbounded solution
C. suggest another optimal basic feasible solution
D. become infeasible

Question No:42 **(Marks:1)** **Vu-Topper RM**
Which of the following is true about an LP problem which is solved by M-method and having infeasible solution?
A. Final iteration table include one of the artificial variables.
B. All choices are equivalent.
C. Penalty 'M' may not able to force artificial variable to zero in final iteration.
D. Associated constraints are not consistent

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Question No:43 **(Marks:1)** **Vu-Topper RM**

If the primal has an unbounded solution, then the dual has------

A. infeasible solution
B. optimal solution
C. bounded solution
D. no solution

Question No:44 **(Marks:1)** **Vu-Topper RM**

Dual of a Dual is ------

A. (Dual)^2
B. Dual
C. Primal
D. Primal-du

Question No:45 **(Marks:1)** **Vu-Topper RM**

In an LP problem, to evaluate the basic variables in Simplex method, which of the following method is used?

A. Gauss Elimination method
B. Gauss-Jordan method
C. Gauss-Seidel method
D. Gauss Least Square method

Question No:46 **(Marks:1)** **Vu-Topper RM**

In two phase method process, first phase _____ the sum of artificial variables.

A. maximize
B. non of these
C. minimize
D. maximize or minimize depending on the situation

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Question No:47 **(Marks:1)** **Vu-Topper RM**

If in the primal problem , there are 2 variables and 3 constraints and the elements of the right hand side of the constraints are 200,567,987 respectively, then its dual objective function is --------.

A. Z = 200y1 + 567y2

B. Z = 200y1 + 567y2 + 987y3

C. Z = y1 + y2 + y3

D. Z = y1 + 3y3

Question No:48 **(Marks:1)** **Vu-Topper RM**

In the initial iteration of Big M-method, the artificial variables appear in ---------.

Non-basic variables' set

Bas

Question No:49 **(Marks:1)** **Vu-Topper RM**

In the final iteration of M-method to obtain the optimal solution, the artificial variable must------- the basis.

A. include in

B. leave

C. add in each variable of

D. subtract from each variable o

Question No:50 **(Marks:1)** **Vu-Topper RM**

A Linear Programming problem may not have any feasible solution.

False

True

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Question No:51 **(Marks:1)** **Vu-Topper RM**

Minimizing Z = 4x1 + 2x2 + x3 is equivalent to max imizing Z = _____.

A. −4x1 − 2x2 − x3
B. −4x1 + 2x2 − x3
C. −4x1 + 2x2 + x3
D. −4x1 − 2x2 + x3

Question No:52 **(Marks:1)** **Vu-Topper RM**

In two phase method, for the phase-I, if the given problem has feasible solution then----------.

A. both objective function and artificial are zero
B. objective function is arbitrary but artificial is zero
C. objective function is zero but artificial may arbitrary
D. both objective and artificial can have arbitrary values

Question No:53 **(Marks:1)** **Vu-Topper RM**

When _____ fail to provide an initial basic feasible solution in the simplex method, we use the Big M Method.

A. slack variables
B. pivot variable
C. artificial variables
D. objective function

Question No:54 **(Marks:1)** **Vu-Topper RM**

When constraints are of greater than or equal to type, then we _____.

A. non of these
B. both of these
C. add slack variable and subtract artificial varialbe

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D. subtract slack variable and add artificial variable

Question No:55 **(Marks:1)** **Vu-Topper RM**

A linear programming problem _____ optimal solutions.

A. must have the multiple
B. may not have the multiple
C. must not have the multiple
D. may have the multiple

Question No:56 **(Marks:1)** **Vu-Topper RM**

If an artificial variable appears in the basis of optimal table in linear programming, what does this indicate about the feasibility of the problem?

A. Feasible optimal solution is bounded
B. Feasible solution is degenerate
C. No feasible optimal solution exists
D. Feasible solution is unbounded

Question No:57 **(Marks:1)** **Vu-Topper RM**

While solving an LP problem, if the candidate for a leaving variable becomes zero, then solution is said to be --------.

A. degenerate solution
B. non-degenerate solution
C. infeasible solution
D. optimal solution

Question No:58 **(Marks:1)** **Vu-Topper RM**

For finding the maximum profit in an enterprise of selling two products such that 'freezing' the sale of one product and keep selling the other. This scenario is studied under ----------.

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A. Duality
B. Artificial variable technique
C. Degeneracy
D. Un-boundedness

Question No:59 **(Marks:1)** **Vu-Topper RM**
In two phase method, for the phase-I, a new objective function is expressed as -----.
A. difference of slack variables
B. sum of artificial variables
C. sum of slack variables
D. difference of artificial variables

Question No:60 **(Marks:1)** **Vu-Topper RM**
Primal of a dual is-----.
A. Prime primal
B. Primal
C. Prime dual
D. Dual

Question No:61 **(Marks:1)** **Vu-Topper RM**
In Simplex method, which of the following is the standard equality corresponding to the constraint"3x+5y>=2"?
A. 3x+2y–S+A=2
B. 3x+2y–S–A=2
C. 3x+2y+S+A=2
D. 3x+2y+S–A=2

Question No:62 **(Marks:1)** **Vu-Topper RM**
When we have type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.
A. artificial, Two phase

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B. artificial, Big M
C. slack, Big M
D. artificial and slack, Big M or Two phase

Question No:63 **(Marks:1)** **Vu-Topper RM**
In the Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column's entries should be --------- except key row(pivot) entry.

A. zero
B. maximum or minimum
C. strictly negative
D. strictly positive

Question No:64 **(Marks:1)** **Vu-Topper RM**
If Primal Problem is a maximization problem, then the dual will be ------

A. mixed Problem
B. maximization Problem
C. none of the above
D. minimization Problem

Question No:65 **(Marks:1)** **Vu-Topper RM**
In a transportation problem, if the number of sources is equal to the number of destinations, how many allocations are needed for a non-degenerate solution?

A. Two less than the number of sources/destinations
B. Equal to the number of sources/destinations
C. Three less than the number of sources/destinations
D. One less than the number of sources/destinations

Question No:66 **(Marks:1)** **Vu-Topper RM**
After allocating and eliminating a fully satisfied row/column, what is the NEXT step in VAM?

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A. Introduce a new row or column to the table.
B. Change all remaining cell costs to zero.
C. Recalculate penalties for the reduced table.
D. Stop the process, as the solution is found.

Question No:67 **(Marks:1)** **Vu-Topper RM**
If multiple rows/columns have the same highest penalty, then which of the following is true?
A. Calculate new penalties for the entire table before allocating.
B. Declare the problem impossible to solve with VAM.
C. Choose one arbitrarily and proceed.
D. Allocate to all of them simultaneously.

Question No:68 **(Marks:1)** **Vu-Topper RM**
For an unbalanced Transportation problem, we have to open a dummy row or dummy column depending on the need, then the unit cost of transportation of cells of dummy row or column will be taken ----------.
A. epsilon
B. -1
C. arbitrary
D. zero

Question No:69 **(Marks:1)** **Vu-Topper RM**
What does the difference between the smallest and next to the smallest element in a row or column represent in VAM?
A. Total cost of transportation
B. Penalty for wrong allocation
C. Maximum feasible amount
D. Unit cost of transportation

Question No:70 **(Marks:1)** **Vu-Topper RM**
The column, which is introduced in the matrix to balance an unbalanced Transportation problem, is known as-------

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A. key column
B. slack column
C. dummy Column
D. idle column

Question No:71 (Marks:1) Vu-Topper RM

In the Vogel's approximation Method for solving a Transportation problem, Penalty measure for any row or column, is given by which of the following?

A. Difference between the largest unit cost to the next largest cost in the same row(column)
B. Difference between any two arbitrary costs in the came row(column)
C. Difference between the Largest unit cost to the smallest unit cost in the came row(column)
D. Difference between the smallest unit cost to the next smallest cost in the same row(column)

Question No:72 (Marks:1) Vu-Topper RM

The North – West Corner Rule

A. Is based on the concept of minimizing opportunity cost.
B. Is used to find an initial feasible solution
C. Is used to find optimal solution
D. None of above

Question No:73 (Marks:1) Vu-Topper RM

How is the maximum feasible quantity determined in the LCM?

A. By calculating the transportation cost
B. By multiplying the demand and supply
C. By identifying the minimum value
D. By selecting the next least cost

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Question No:74 **(Marks:1)** **Vu-Topper RM**

Which values of "aij's" in the standard form of a linear programming problem will transform it in standard Transportation problem?

A. aij = 0 i = 1, 2,⋯, m; j = 1, 2,⋯, n
B. aij = 0 i = 1, 2,⋯, m; j = 1, 2,⋯, n
C. aij = 0 i = 1, 2,⋯, m; j = 1, 2,⋯, n
D. aij = 1 i = 1, 2,⋯, m; j = 1, 2,⋯, n

Question No:75 **(Marks:1)** **Vu-Topper RM**

In which of the following method to solve a Transportation problem, we have to calculate one more row and one more column?

A. North West Corner method
B. Least Cost Method
C. Vogel's approximation Method
D. Simplex Method

Question No:76 **(Marks:1)** **Vu-Topper RM**

In North West Corner method, in the first row and first column, available resource and sink contain '5' and '7' units respectively; then which of the following amount 'x11' will be allocated in the cell (1,1)?

A. Average(5,7) = 6
B. Min(5,7) = 5
C. Min(5,7-5) = 2
D. Max(5,7) = 7

Question No:77 **(Marks:1)** **Vu-Topper RM**

In which of the following method to solve a Transportation problem, allocation in a cell depends upon penalty cost?

A. Simplex Method
B. Vogel's approximation Method
C. Least Cost Method
D. North West Corner Method

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Vu Topper RM

Question No:78 **(Marks:1)** **Vu-Topper RM**

For North West Corner method, in the first row and first column, resource and sink contain '5' and '7' units respectively; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

A. (2,2)
B. (5,2)
C. (1,2)
D. (2,1)

Question No:79 **(Marks:1)** **Vu-Topper RM**

What is the penalty in Vogel's Approximation Method (VAM)?

A. The amount allocated to each cell in the transportation table
B. The number of iterations required to find the optimal solution
C. The total transportation cost
D. The difference between the smallest and next to the smallest element in a row or column

Question No:80 **(Marks:1)** **Vu-Topper RM**

When the total allocations in a transportation model of m*n size is not equals to "m + n – 1", the situation is known as:

A. Degeneracy
B. Unbalanced situation
C. None of the above
D. Tie situation

Question No:81 **(Marks:1)** **Vu-Topper RM**

The objective of a transportation problem is to develop an ---------- transportation schedule that meets all demand from given stock at a ------ total shipping cost.

A. (fractional, maximum)
B. (integral, minimum)
C. (fractional, minimum)

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D. (integral, maximum)

Question No:82 **(Marks:1)** **Vu-Topper RM**

If the total demand is equal to total supply as per requirement of a balanced transportation problem i-e " " then
which of the following is true?

A. "a1=b1,a2=b2,⋯,an=bn" is NOT necessarily implied

B. " a1=b1,a2=b2,⋯,an=bn " is necessarily implied

Question No:83 **(Marks:1)** **Vu-Topper RM**

Which method is expected to give a better initial solution among NWC, LCM, and VAM?

A. VAM

B. NWC

C. All methods provide similar initial solutions

D. LCM

Question No:84 **(Marks:1)** **Vu-Topper RM**

Which of the following Method is comparatively little complex to solve a Transportation problem?

A. Least Cost Method

B. Vogel's approximation Method

C. Hungarian's Method

D. North West Corner Method

Question No:85 **(Marks:1)** **Vu-Topper RM**

How is the least cost method (LCM) different from the Northwest Corner rule (NWC)?

A. LCM follows a random allocation strategy

B. LCM allocates the maximum feasible amount

C. LCM considers unit costs of transportation

D. LCM starts from the bottom right corner

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Question No:86 **(Marks:1)** **Vu-Topper RM**

What causes degeneracy in a transportation problem?

A. Less than (m + n - 1) occupied cells
B. Unbalanced supply/demand
C. No feasible solution
D. Multiple optimal solutions

Question No:87 **(Marks:1)** **Vu-Topper RM**

Which of the following can be said the main objective of the Least Cost Method to solve a transportation problem?

A. Concentrating on the routes with least transportation time
B. Concentrating on the routes with expensive cost of transportation
C. Concentrating on the routes with maximum available transportation time
D. Concentrating on the routes with cheapest cost of transportation

Question No:88 **(Marks:1)** **Vu-Topper RM**

You realize that a row's total supply cannot fully satisfy the demand in any column. This means that_____.

A. You must add transportation costs to increase supply.
B. The original transportation problem has no feasible solution.
C. You can ignore this row and continue.
D. You have made an error in earlier allocations.

Question No:89 **(Marks:1)** **Vu-Topper RM**

The transportation problem can be represented mathematically as a linear programming model with the objective function of _____.

A. Maximizing the total transportation cost
B. Minimizing the total transportation cost
C. Maximizing the total production cost
D. Minimizing the total profit

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Question No:90 **(Marks:1)** **Vu-Topper RM**

After identifying the row with the least cost, what action is taken in the LCM?

A. Determine the next least cost
B. Repeat the process for all columns
C. Eliminate the row
D. Allocate the maximum feasible quantity

Question No:91 **(Marks:1)** **Vu-Topper RM**

The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if numbers of positive allocations are

A. m+n
B. m-n+1
C. m+n+1
D. m+n-1

Question No:92 **(Marks:1)** **Vu-Topper RM**

In the LCM, what is done if the minimum cost is not unique?

A. Break the tie arbitrarily
B. Choose the next least cost
C. Repeat the process with a new table
D. Increase the cost arbitrarily

Question No:93 **(Marks:1)** **Vu-Topper RM**

While solving a Transportation problem by Vogel's approximation method, if the row has two elements of the same magnitude say 'x' as the smallest element then the opportunity(penalty) cost of that row or column is ----------.

A. x
B. 2x
C. –x
D. Zero

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Question No:94 **(Marks:1)** **Vu-Topper RM**

Transportation problem is basically a --------

A. Iconic model
B. Maximization model
C. Transshipment problem
D. Minimization model

Question No:95 **(Marks:1)** **Vu-Topper RM**

What is the key characteristic of Vogel's Approximation Method (VAM)?

A. It provides a slightly higher-cost initial solution.
B. It works for any size of transportation problem.
C. It always finds the best possible (optimal) solution.
D. It requires fewer computational steps.

Question No:96 **(Marks:1)** **Vu-Topper RM**

In the Least Cost Method, allocations are made in a cell by assigning ----------to cell with smallest unit cost.

A. any arbitrary amount
B. minimum
C. average of maximum and minimum
D. maximum

Question No:97 **(Marks:1)** **Vu-Topper RM**

If a Transportation problem has ‘m+n’ constraints then how many number of unknowns it will have?

A. m.n
B. m/n
C. m+n
D. m–n

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Question No:98 **(Marks:1)** **Vu-Topper RM**

What is the significance of selecting the row or column with the largest penalty in VAM?

A. It represents the maximum supply or demand
B. It prioritizes the allocation to minimize penalties
C. It indicates the highest transportation cost
D. It denotes the final allocation step

Question No:99 **(Marks:1)** **Vu-Topper RM**

The Transportation and Transshipment Models are types of ---------- models.

A. Network
B. Variation
C. Allocation
D. Inventory

Question No:100 **(Marks:1)** **Vu-Topper RM**

When does degeneracy occur in transportation problems?

A. When there are negative allocations in the initial feasible solution.
B. When there are excess allocations in the initial feasible solution.
C. When there are no allocations in the initial feasible solution.
D. When there are fewer allocations than required in the initial feasible solution.

Question No:101 **(Marks:1)** **Vu-Topper RM**

The row, which is introduced in the matrix to balance an unbalnced Transportation problem, is known as ----------.

A. dummy row
B. key row
C. slack row
D. idle row

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Question No:102 **(Marks:1)** **Vu-Topper RM**

What is Vogel's Approximation Method (VAM) based on?

A. Greedy strategy

B. Considering only the maximum feasible amount

C. Random allocation of resources

D. Calculating penalties for wrong allocations

Question No:103 **(Marks:1)** **Vu-Topper RM**

What is the key limitation of Vogel's Approximation Method?

A. Doesn't always find the absolute best solution.

B. Requires a balanced supply and demand.

C. It is computationally very slow compared to other methods.

D. Only works on small transportation problems.

Question No:104 **(Marks:1)** **Vu-Topper RM**

For an unbalanced Transportation problem, if the total demand is LESS than total supply then which of the following is true in order to balance the problem?

A. A dummy sink would have to include with demand equal to the surplus

B. A dummy source would have to include with supply equal to shortage

C. One constraint will have to add

D. One constraint will have evacuate

Question No:105 **(Marks:1)** **Vu-Topper RM**

Under which of the following condition in the Least Cost Method, after first allocation in cell (1,1), next allocation along the diagonal in cell (2,2) can be made?

A. Total Supply = Total Demand

B. If after crossed out 1st row or 1st column, cell (2,2) is of minimum cost

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C. If in the second row and second column, each resource and sink contain equal units
D. If Total Supply is not equal to Total Demand

Question No:106 **(Marks:1)** **Vu-Topper RM**

For a Transportation problem, after finding the initial basic feasible solution, which of the following method is used to find the optimum solution?

A. None of these
B. North West Corner Method
C. Least Cost Method
D. Vogel's approximation Method

Question No:107 **(Marks:1)** **Vu-Topper RM**

Why do we use specialized algorithms for transportation problems instead of the simplex method?

A. The simplex method is used for maximization only.
B. Specialized algorithms are more efficient.
C. Transportation models are too complex.
D. The simplex method cannot handle equality constraints.

Question No:108 **(Marks:1)** **Vu-Topper RM**

During the substitution process of determining the Transportation Cost 'xij' in a balanced Transportation problem, no---------- operation is used as the coefficients of 'xij' are '0' or '1'. Due to this property, every basic feasible solution have only non-negative integral values.

Addition
Division

Question No:109 **(Marks:1)** **Vu-Topper RM**

Which of the following method is used to find the initial basic feasible solution of a transportation problem?

A. Least Cost's Method

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B. Vogel's approximation Method
C. North West Corner Method
D. All above methods are applicable

Question No:110 (Marks:1) Vu-Topper RM
In the NWC method, which cell is chosen for allocation initially?
A. Top right corner
B. Bottom left corner
C. Bottom right corner
D. Top left corner (Northwest Corner)

Question No:111 (Marks:1) Vu-Topper RM
In North West Corner method, the first step after choosing the appropriate cell in 1st row, we allocate ----------so that the capacity of first row or first column is exhausted.
A. non- negative quantity
B. as least as possible
C. any arbitrary quantity
D. as much as possible

Question No:112 (Marks:1) Vu-Topper RM
Which step involves eliminating fully allocated rows or columns in the least cost method (LCM)?
A. Step 4
B. Step 1
C. Step 2
D. Step 3

Question No:113 (Marks:1) Vu-Topper RM
A balanced transportation model with '5'number of sources and '7' destinations has -------- number of constraint equations.
A. 12
B. 2

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C. 35
D. 10

Question No:114 **(Marks:1)** **Vu-Topper RM**
What action is taken if a row or column is empty in the LCM?
A. Repeat the allocation process
B. Increase the transportation cost
C. Allocate the remaining quantities
D. Eliminate the row or column

Question No:115 **(Marks:1)** **Vu-Topper RM**
In order to test a solution for optimality in a transportation problem, it must have:
A. Zero cost for all non-allocated cells.
B. The same number of supply and demand points.
C. At least one allocation in every row and column.
D. Exactly (m + n - 1) occupied cells.

Question No:116 **(Marks:1)** **Vu-Topper RM**
For a Transportation Problem, if it's initial feasible solution is evaluated by Least Cost Method, then quality of this initial solution is better than ---------.
North West Corner Method
Vogal's approximation Method

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